

Student:

James Collins

Email:

collins222@usf.edu

Time on Task:

1 hour, 10 minutes

Progress:

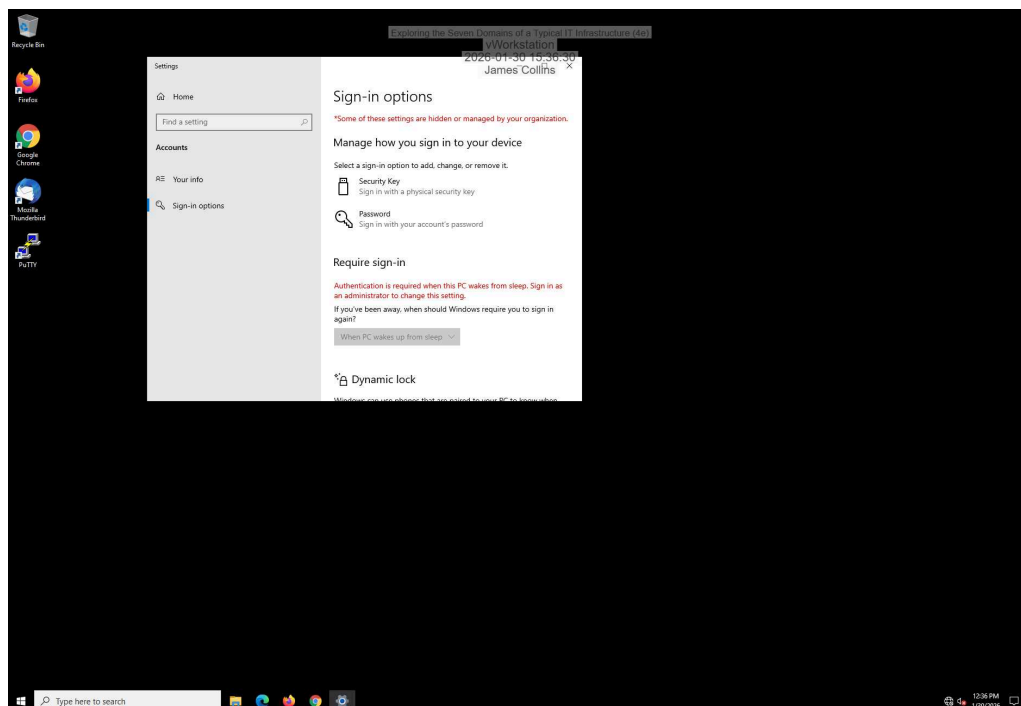
94%

Report Generated: Friday, January 30, 2026 at 4:37 PM

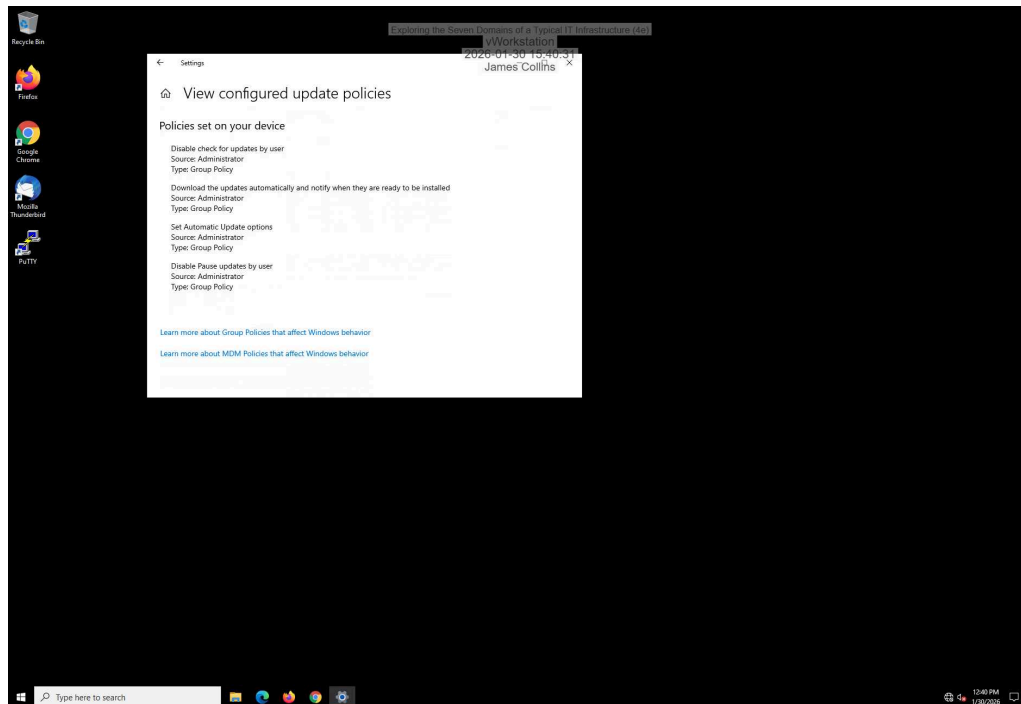
Section 1: Hands-On Demonstration

Part 1: Explore the Workstation Domain

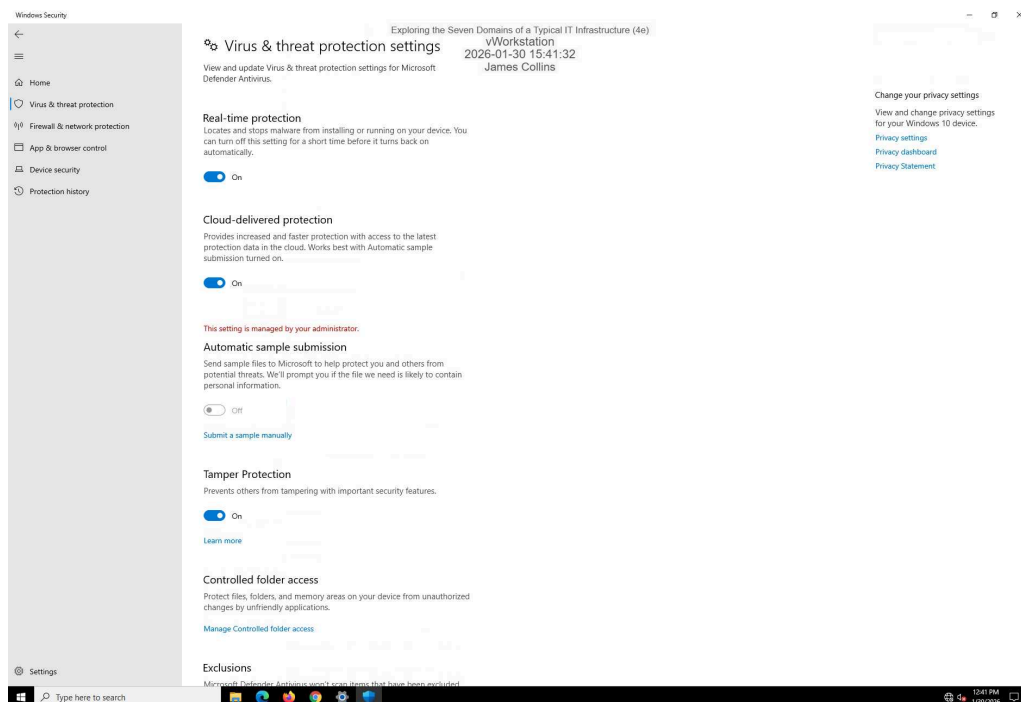
4. Make screen capture showing the **Sign-in options** for Alice's account.



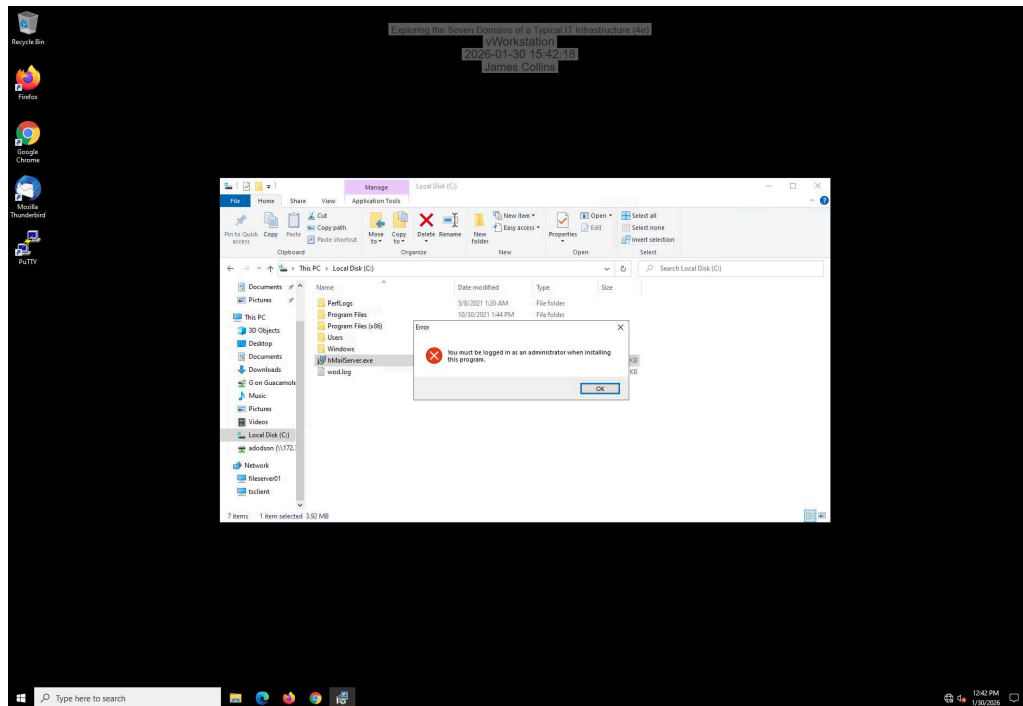
7. Make a screen capture showing the View configured update policies page.



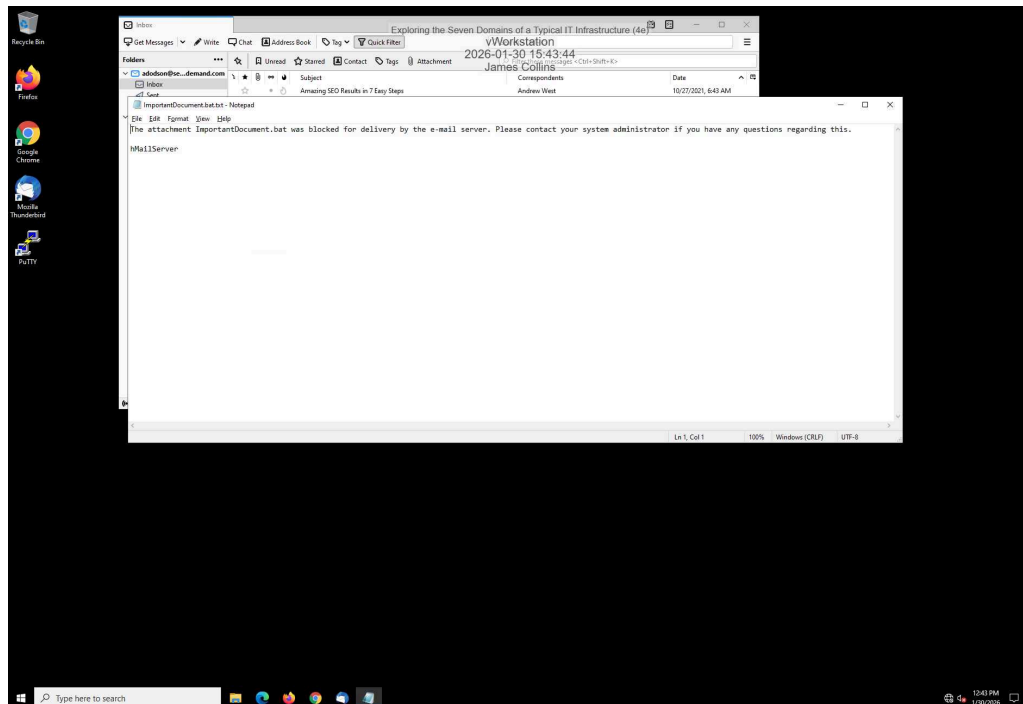
14. Make a screen capture showing the Virus & Threat Protection Settings.



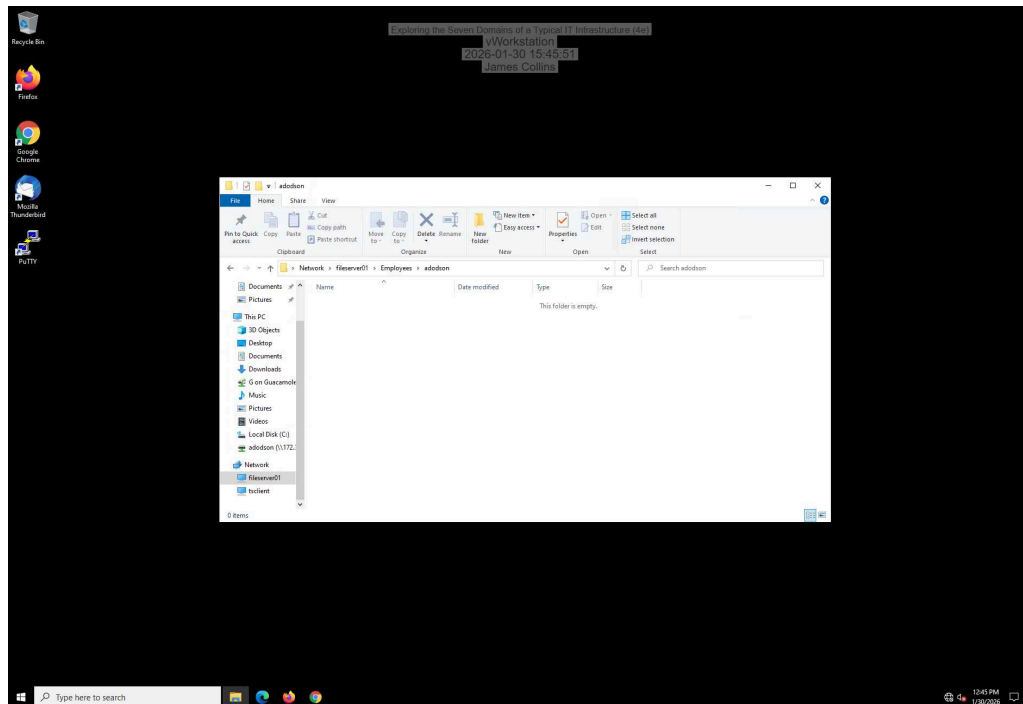
18. Make a screen capture showing the security warning from attempting to run an executable file.



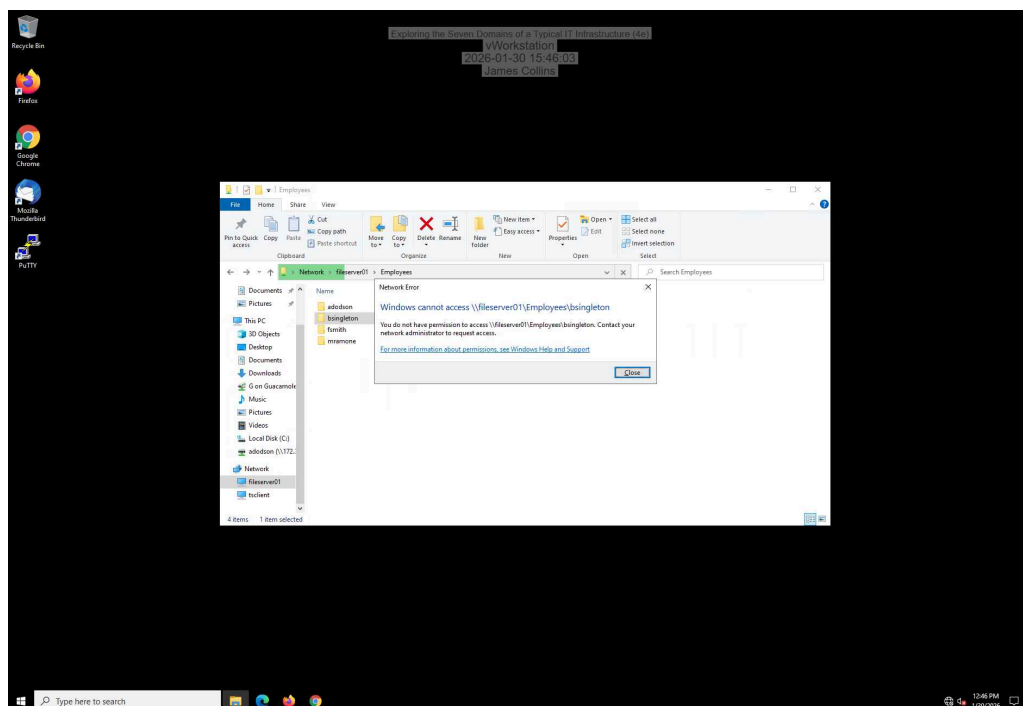
24. Make a screen capture showing the blocked attachment message.



28. Make a screen capture showing a successful connection to the adodson user folder.



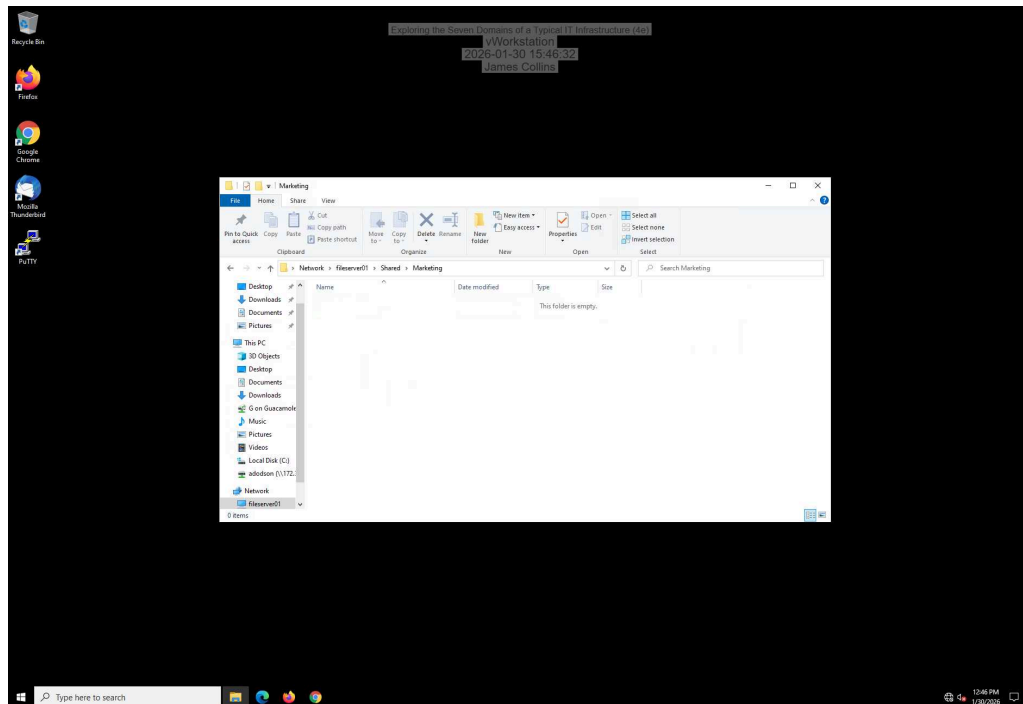
29. Make a screen capture showing a failed connection to another user folder.



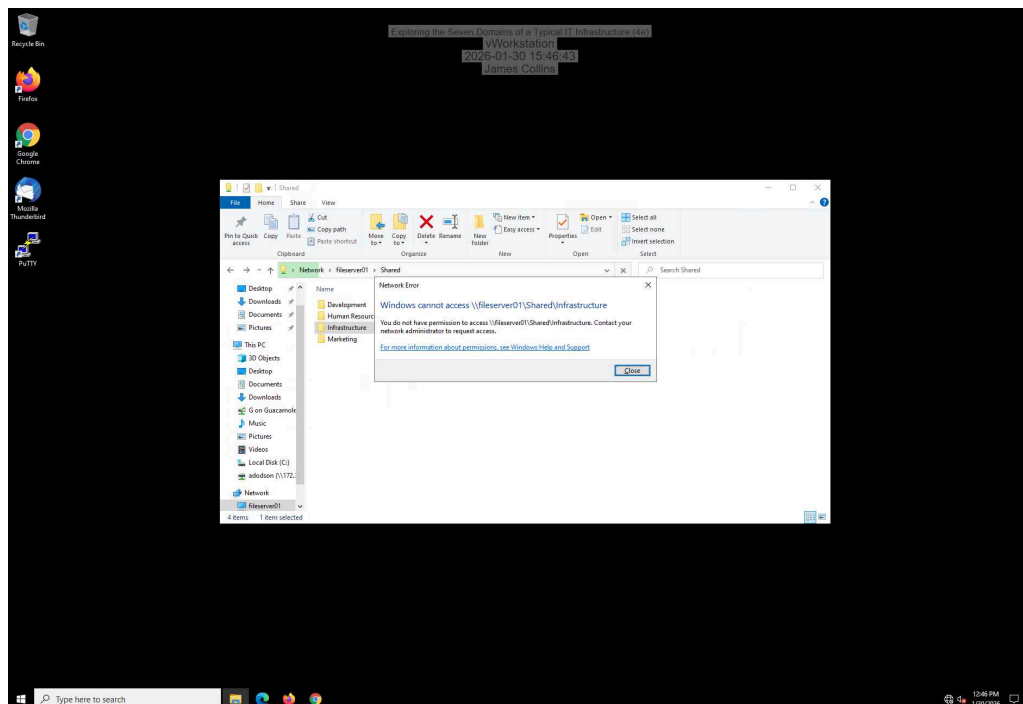
Exploring the Seven Domains of a Typical IT Infrastructure (4e)

Fundamentals of Information Systems Security, Fourth Edition - Lab 01

31. Make a screen capture showing a successful connection to the Marketing shared folder.

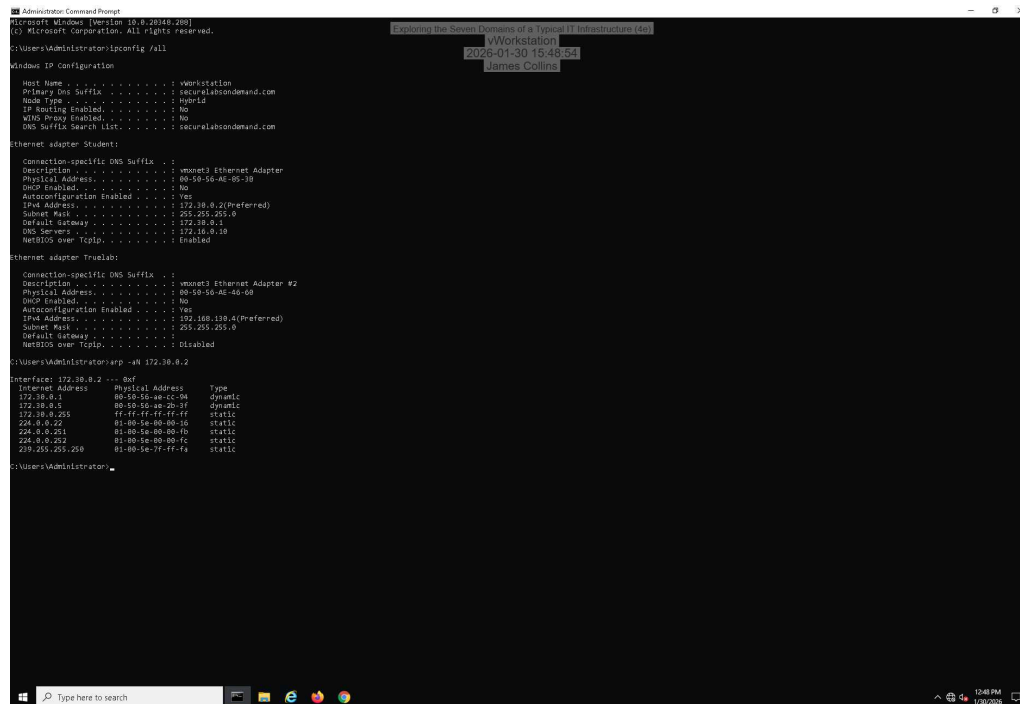


32. Make a screen capture showing a failed connection to another shared folder.



Part 2: Explore the LAN Domain

5. Make a screen capture showing the vWorkstation's original ARP table.



The screenshot shows a Windows Command Prompt window with the following text:

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20H4-200]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>ipconfig /all

Windows IP Configuration

Host Name . . . . . : vWorkstation
Primary DNS Suffix . . . . . : securelabsondemand.com
Node Type . . . . . : Hybrid
IP Routing Enabled . . . . . : No
WINS Proxy Enabled . . . . . : No
DNS Suffix Search List. . . . . : securelabsondemand.com

Ethernet adapter Student:

Connection-specific DNS Suffix . : 
Description . . . . . : {muonet3} Ethernet Adapter
Physical Address. . . . . : 80-50-56-A6-05-30
Dhcp Enabled . . . . . : No
Autocconfiguration Enabled . . . . : Yes
IP Address . . . . . : 172.30.0.2
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 172.30.0.1
DNS Servers . . . . . : 172.16.0.10
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter TrueLab:

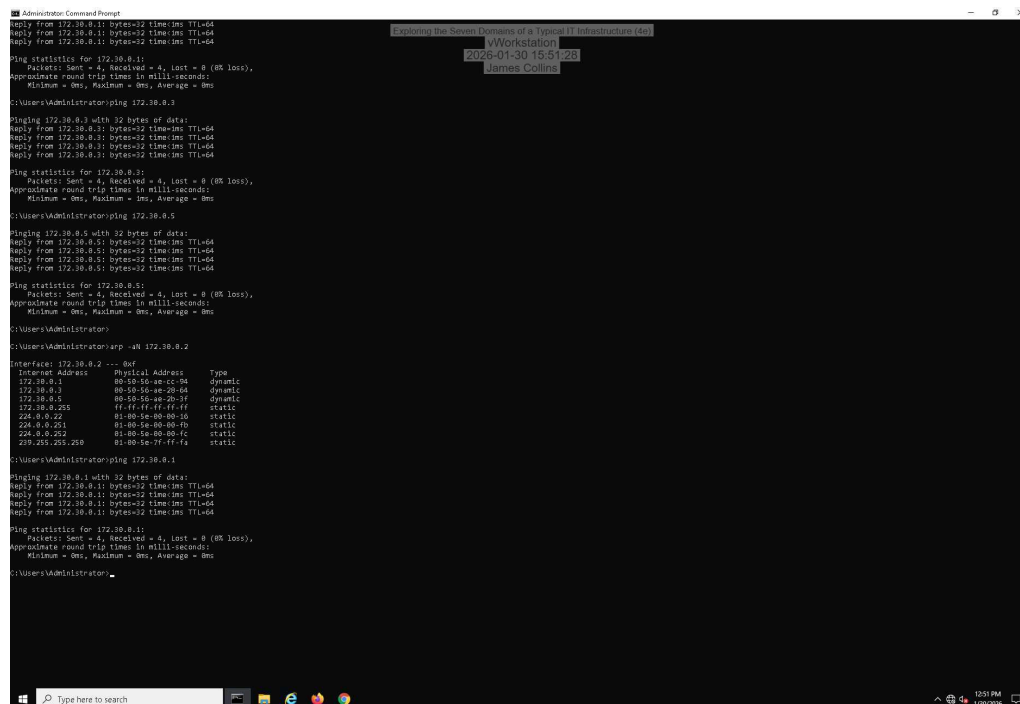
Connection-specific DNS Suffix . : 
Description . . . . . : {muonet3} Ethernet Adapter #2
Physical Address. . . . . : 80-50-56-A6-06-00
Dhcp Enabled . . . . . : No
Autocconfiguration Enabled . . . . : Yes
IP Address . . . . . : 192.168.130.4(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 
DNS Servers . . . . . : 
NetBIOS over Tcpip. . . . . : Disabled

C:\Users\Administrator>arp -a 172.30.0.2

Interface: 172.30.0.2 --- 0x0
Internet Address      Physical Address      Type
172.30.0.1            80-50-56-ae-c0-9a     dynamic
172.30.0.5            80-50-56-ae-2b-3f     dynamic
172.30.0.255          ff-ff-ff-ff-ff-ff     static
224.0.0.251           01-00-5e-00-00-00     static
224.0.0.252           01-00-5e-00-00-00     static
224.0.0.252           01-00-5e-00-00-00     static
259.255.255.250       01-00-5e-7f-ff-fa     static

C:\Users\Administrator>
```

10. Make a screen capture showing the vWorkstation's updated ARP table.



The screenshot shows a Windows Command Prompt window with the following text:

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20H4-200]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>ping 172.30.0.1
Pinging 172.30.0.1: bytes=32 time=1ms TTL=64
Reply from 172.30.0.1: bytes=32 time=1ms TTL=64
Reply from 172.30.0.1: bytes=32 time=1ms TTL=64

Ping statistics for 172.30.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Administrator>ping 172.30.0.3
Pinging 172.30.0.3 with 32 bytes of data:
Reply from 172.30.0.3: bytes=32 time=1ms TTL=64
Reply from 172.30.0.3: bytes=32 time=1ms TTL=64
Reply from 172.30.0.3: bytes=32 time=1ms TTL=64
Reply from 172.30.0.3: bytes=32 time=1ms TTL=64

Ping statistics for 172.30.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Administrator>ping 172.30.0.5
Pinging 172.30.0.5 with 32 bytes of data:
Reply from 172.30.0.5: bytes=32 time=1ms TTL=64
Reply from 172.30.0.5: bytes=32 time=1ms TTL=64
Reply from 172.30.0.5: bytes=32 time=1ms TTL=64
Reply from 172.30.0.5: bytes=32 time=1ms TTL=64

Ping statistics for 172.30.0.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Administrator>arp -a 172.30.0.2

Interface: 172.30.0.2 --- 0x0
Internet Address      Physical Address      Type
172.30.0.1            80-50-56-ae-c0-9a     dynamic
172.30.0.3            80-50-56-ae-2b-3f     dynamic
172.30.0.5            80-50-56-ae-2b-3f     dynamic
172.30.0.255          ff-ff-ff-ff-ff-ff     static
224.0.0.251           01-00-5e-00-00-00     static
224.0.0.252           01-00-5e-00-00-00     static
224.0.0.252           01-00-5e-00-00-00     static
259.255.255.250       01-00-5e-7f-ff-fa     static

C:\Users\Administrator>ping 172.30.0.1
Pinging 172.30.0.1 with 32 bytes of data:
Reply from 172.30.0.1: bytes=32 time=1ms TTL=64
Reply from 172.30.0.1: bytes=32 time=1ms TTL=64
Reply from 172.30.0.1: bytes=32 time=1ms TTL=64
Reply from 172.30.0.1: bytes=32 time=1ms TTL=64

Ping statistics for 172.30.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Administrator>
```

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[illegible]

```
172.30.25 -PUTY
Last login: Thu Oct 26 10:19:07 2021 from 172.30.25.2
FreeBSD 12.2-RELEASE-pid 40474eb984 (MBA) VTOEMAS

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http://truenas.com

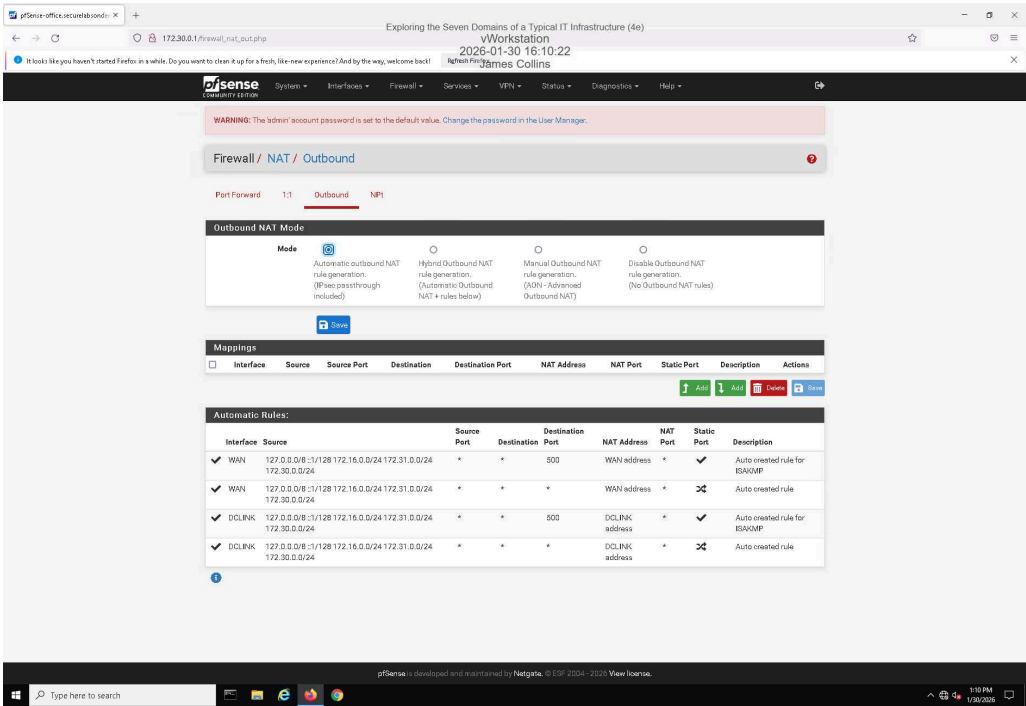
Welcome to TrueNAS

WARNING! Settings changed through the CLI are not written to
the configuration database and will be reset on reboot.

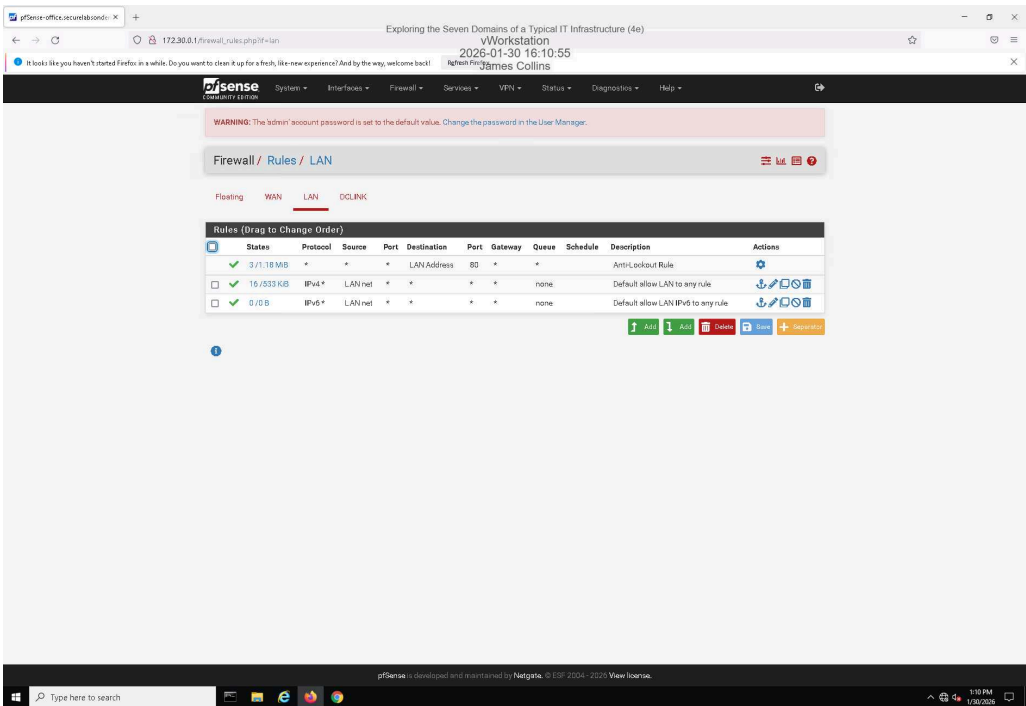
root@fileserver01:~# pwd
/root
root@fileserver01:~# whoami
jlsassot
root@fileserver01:~# whoamiend /usr
who: command not found; whoamiend
root@fileserver01:~# whoami
root
root@fileserver01:~# cd /usr
root@fileserver01:/usr# ls -l
total 5
drwxr-xr-x 1 root wheel 5 Oct 12 2021 md_etc
drwxr-xr-x 4 root wheel 4 Oct 28 2021 tank
root@fileserver01:/usr# ls -la tank
Employees shared
root@fileserver01:/usr# cd Employees
pwd: no such file or directory: Employees
root@fileserver01:/usr# cd ..
root@fileserver01:/usr# ls
authz      cmd       boot     compset  conf      data     entropy  home    libexec  net      proc     root     storage.log  usr
        end         dir             COPYRIGHT  dev              etc               lib                media          nss            rcache      root      shn                 storcli.log   var
root@fileserver01:/usr# ls -l
total 48
drwxr-xr-x 2 root wheel 1023 Oct 12 2021 .sshrc
drwxr-xr-x 2 root wheel 405 Oct 12 2021 .profile
drwxr-xr-x 1 root wheel 13 Oct 12 2021 .rnd => /var/tmp/.rnd
drwxr-xr-x 2 root wheel 47 Oct 12 2021 bin
drwxr-xr-x 10 root wheel 67 Jan 24 2020 boot
drwxr-xr-x 3 root wheel 3 Oct 12 2021 compset
drwxr-xr-x 3 root wheel 3 Oct 12 2021 conf
drwxr-xr-x 1 root wheel 4113 Oct 12 2021 COPYRIGHT
drwxr-xr-x 6 root wheel 13 Jan 30 2019 dmca
drwxr-xr-x 210 root wheel 512 Jan 30 2019 dev
drwxr-xr-x 1 root wheel 4098 Jan 30 2019 entropy
drwxr-xr-x 32 root wheel 6960 Jan 30 2019 etc
drwxr-xr-x 3 root wheel 3 Oct 12 2021 home
drwxr-xr-x 5 root wheel 58 Oct 12 2021 lib
drwxr-xr-x 3 root wheel 4 Oct 12 2021 libexec
drwxr-xr-x 2 root wheel 2 Oct 12 2021 libmedia
drwxr-xr-x 3 root wheel 128 Jan 30 2019 nss
drwxr-xr-x 2 root wheel 2 Oct 12 2021 nss
drwxr-xr-x 2 root wheel 2 Oct 12 2021 proc
drwxr-xr-x 2 root wheel 145 Oct 12 2021 rcache
drwxr-xr-x 5 root wheel 16 Jan 30 2019 root
drwxr-xr-x 2 root wheel 131 Oct 12 2021 shell
drwxr-xr-x 1 root wheel 2057 Oct 12 2021 sshrc
drwxr-xr-x 1 root wheel 8 Oct 12 2021 tmp => /var/tmp
drwxr-xr-x 13 root wheel 13 Oct 12 2021 usr
drwxr-xr-x 28 root wheel 170 Jan 30 2019 vnc
root@fileserver01:/usr# cd tank/Employees
pwd: no such file or directory: tank/Employees
root@fileserver01:/usr# cd ../usr
root@fileserver01:/usr# ls
md_etc tank
root@fileserver01:/usr# cd tank
root@fileserver01:/usr/tank# ls
Employees shared
root@fileserver01:/usr/tank# cd Employees
root@fileserver01:/usr/tank/Employees# ls
authentication  email
root@fileserver01:/usr/tank/Employees#
```

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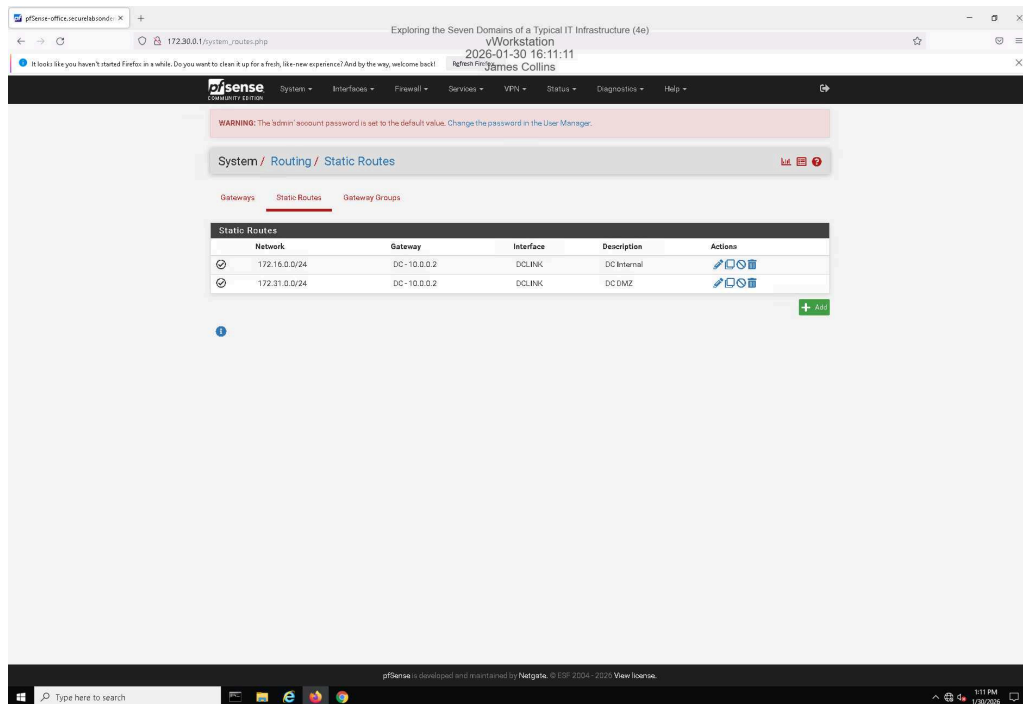
6. Make a screen capture showing the Outbound NAT settings.



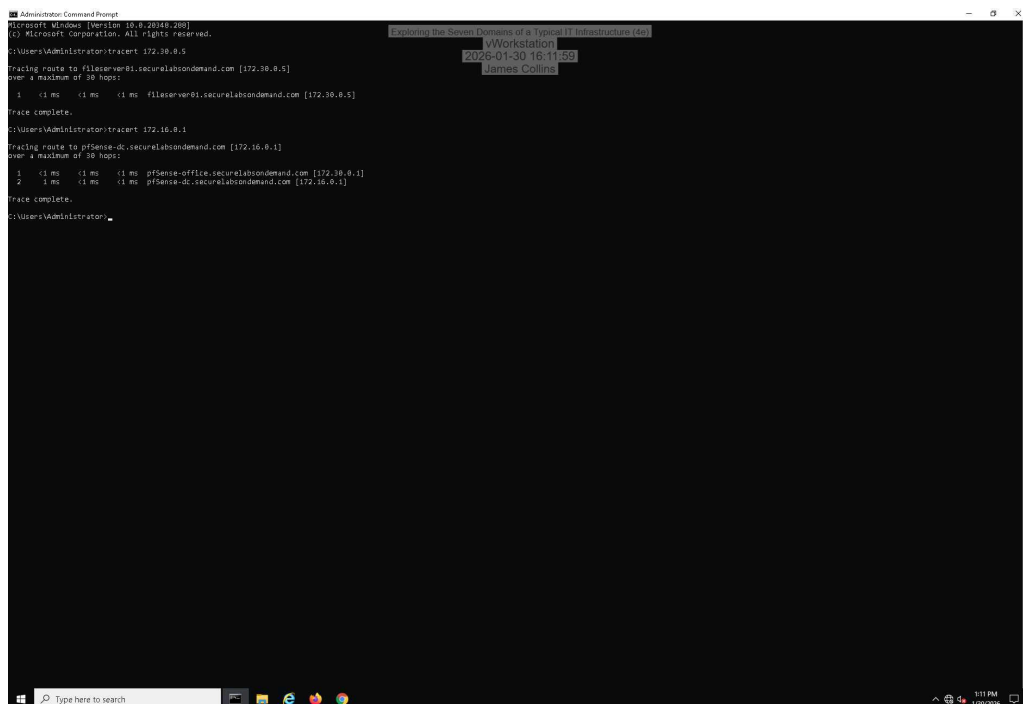
9. Make a screen capture showing the permissive LAN rules.



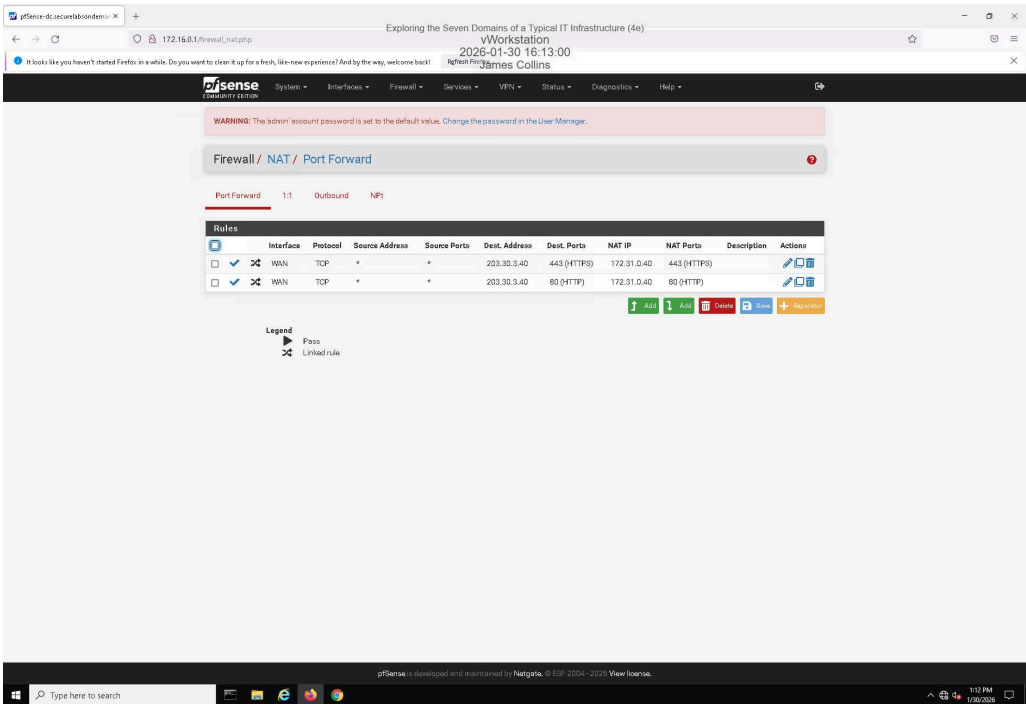
12. Make a screen capture showing the **Static Routes** page.



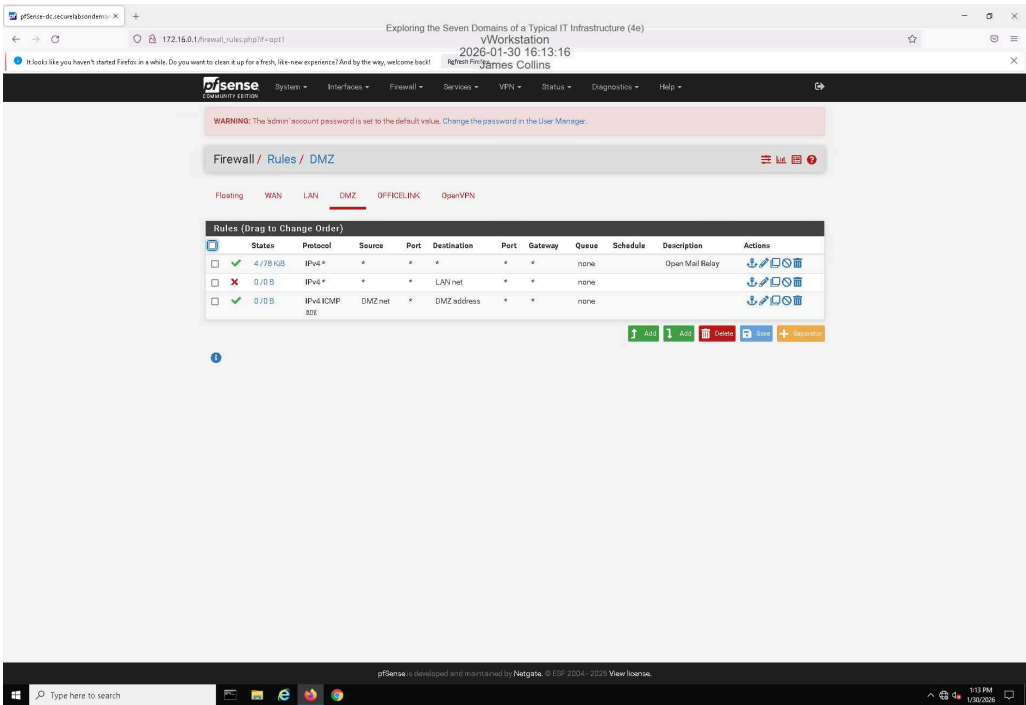
16. Make a screen capture showing the **result of your tracer** to the pfsense-dc appliance.



22. Make a screen capture showing the **Port Forward** rules for the web server.



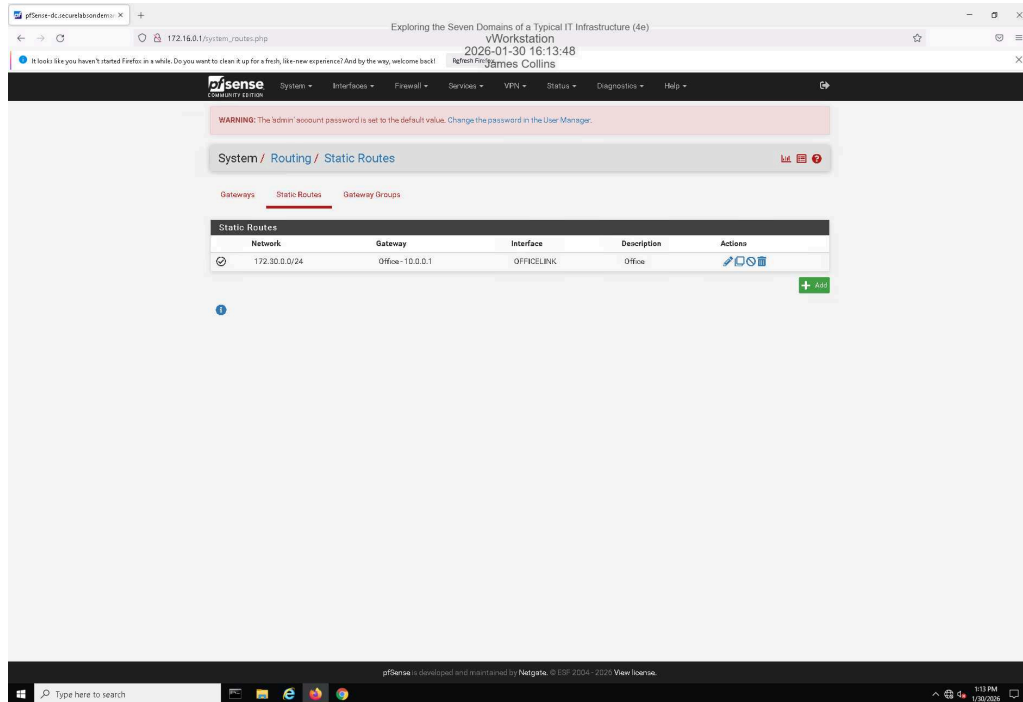
25. Make a screen capture showing the **DMZ** firewall rules.



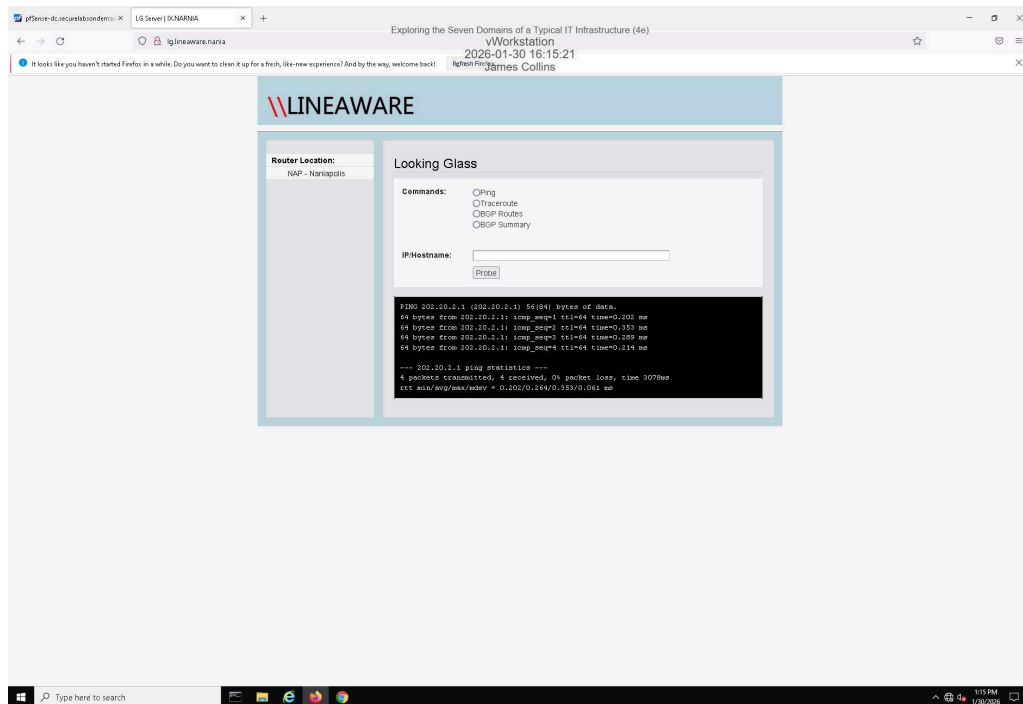
Section 2: Applied Learning

Part 1: Explore the WAN Domain

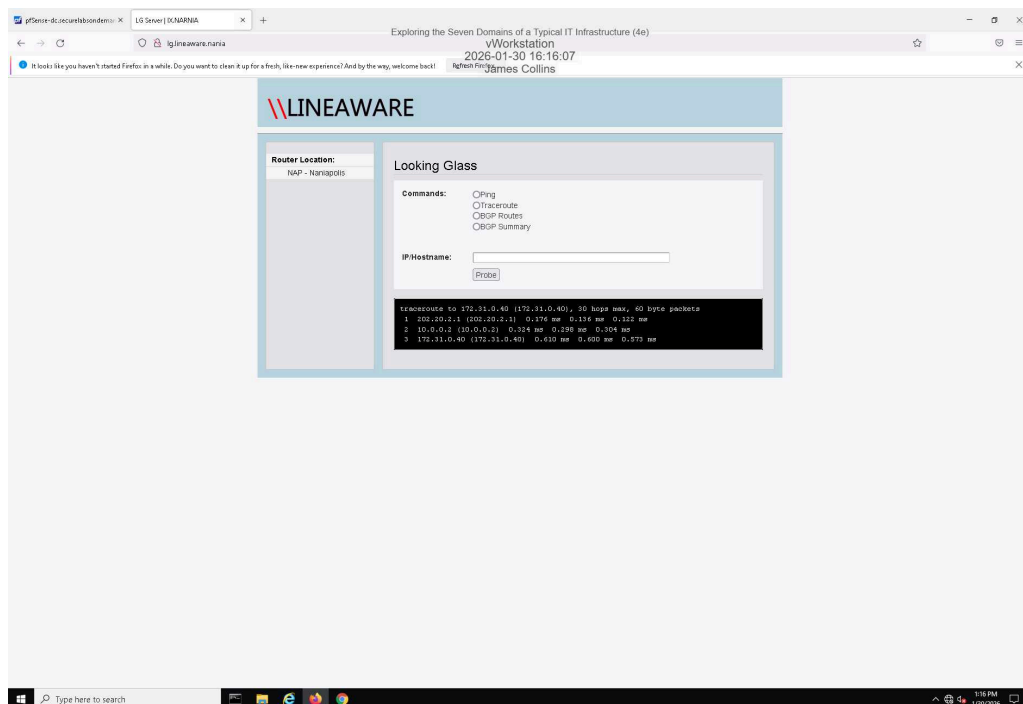
5. Make a screen capture showing the **static route** for the point-to-point connection.



9. Make a screen capture showing the **BPG neighbor ping** results.

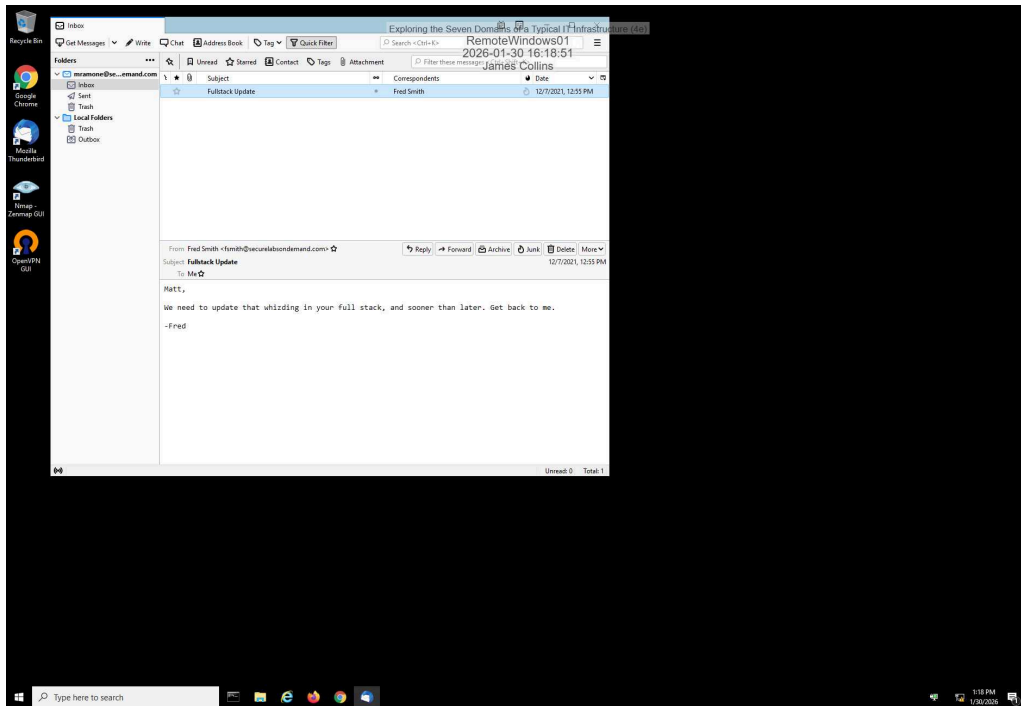


12. Make a screen capture showing the **traceroute** to the file server.



Part 2: Explore the Remote Access Domain

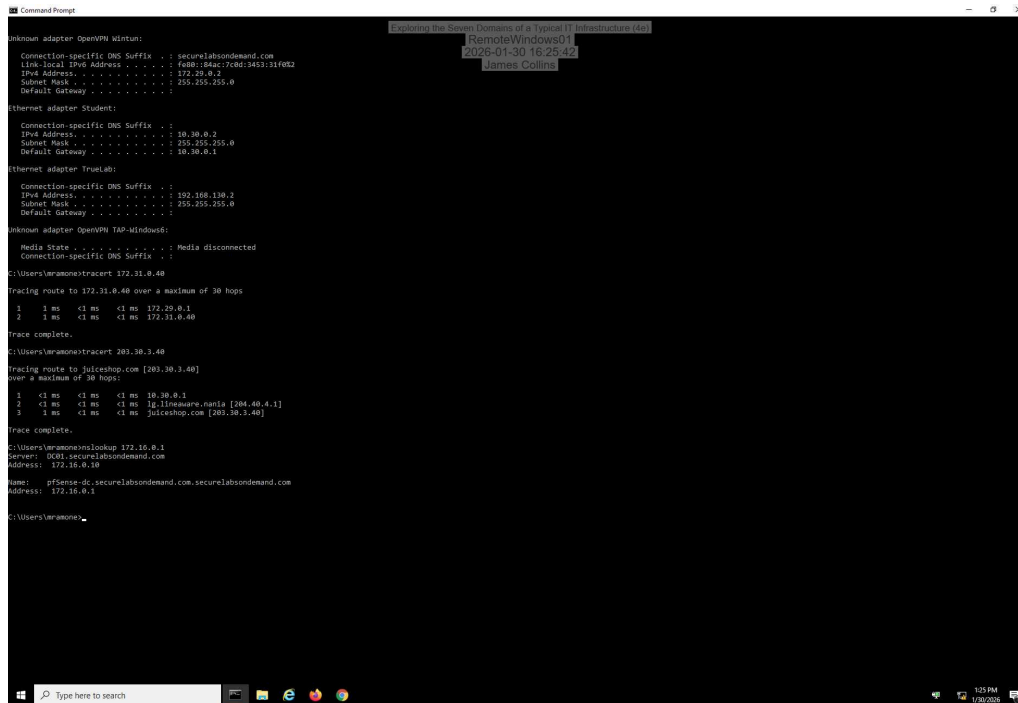
9. Make a screen capture showing the **successful connection to the email server**.



14. **Document** whether the VPN connection is split tunnel or full tunnel, based on the tracert results.

split tunnel

16. Make a screen capture showing the successful reverse DNS lookup for the internal host.



```
C:\Users\james>ipconfig /all

Ethernet adapter OpenVPN Wintun:

   Connection-specific DNS Suffix  . : securelabsondemand.com
   Link-local IPv6 Address . . . . . : fe80::84ac:7c0d:3453:31f0a2
   IPv4 Address. . . . . : 172.29.0.2
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . :

Ethernet adapter Student:

   Connection-specific DNS Suffix  . : 
   IPv4 Address. . . . . : 10.30.0.2
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . : 10.30.0.1

Ethernet adapter TrueLab:

   Connection-specific DNS Suffix  . : 
   IPv4 Address. . . . . : 192.168.130.2
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . :

Unknown adapter OpenVPN TAP-Windows:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix  . : 

C:\Users\james>tracert 172.31.0.40

Tracing route to 172.31.0.40 over a maximum of 30 hops:
  0  1 ms  <1 ms  <1 ms  172.29.0.1
  1  1 ms  <1 ms  <1 ms  172.31.0.40
Trace complete.

C:\Users\james>tracert 203.30.3.40

Tracing route to juiceshop.com [203.30.3.40]
over a maximum of 30 hops:
  0  <1 ms  <1 ms  <1 ms  10.30.0.1
  1  <1 ms  <1 ms  <1 ms  10.111.111.111 [204.40.4.1]
  2  1 ms  <1 ms  <1 ms  juiceshop.com [203.30.3.40]
Trace complete.

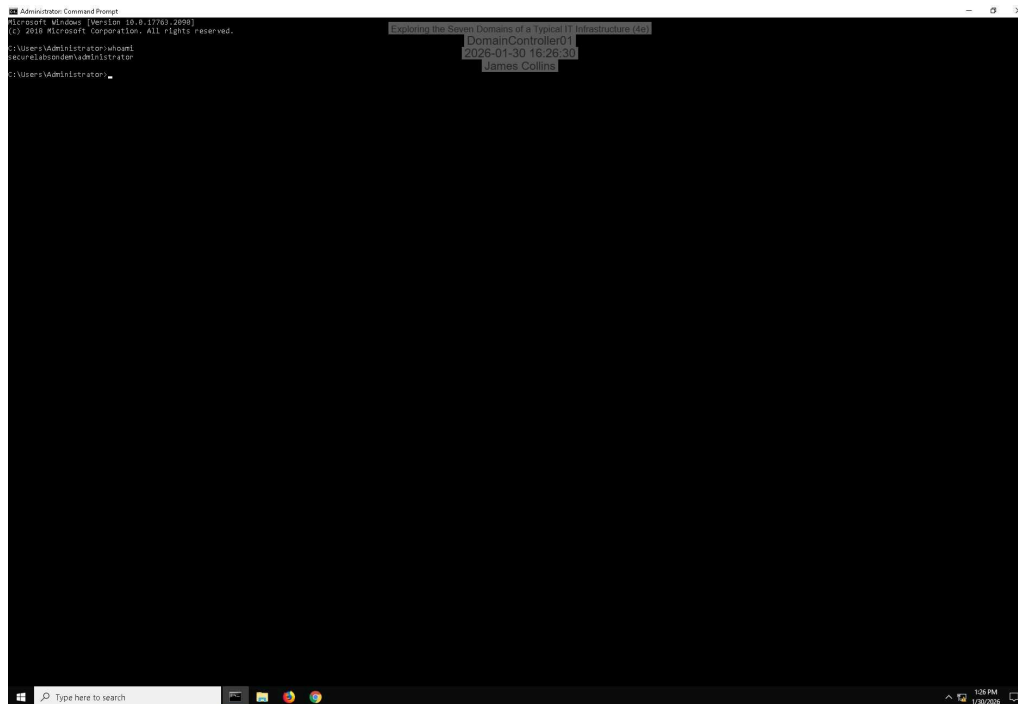
C:\Users\james>nslookup 172.10.0.1
Server: 10.30.0.1, securelabsondemand.com
Address: 172.10.0.1

Name:    pfsense-dc.securelabsondemand.com.securelabsondemand.com
Address: 172.10.0.1

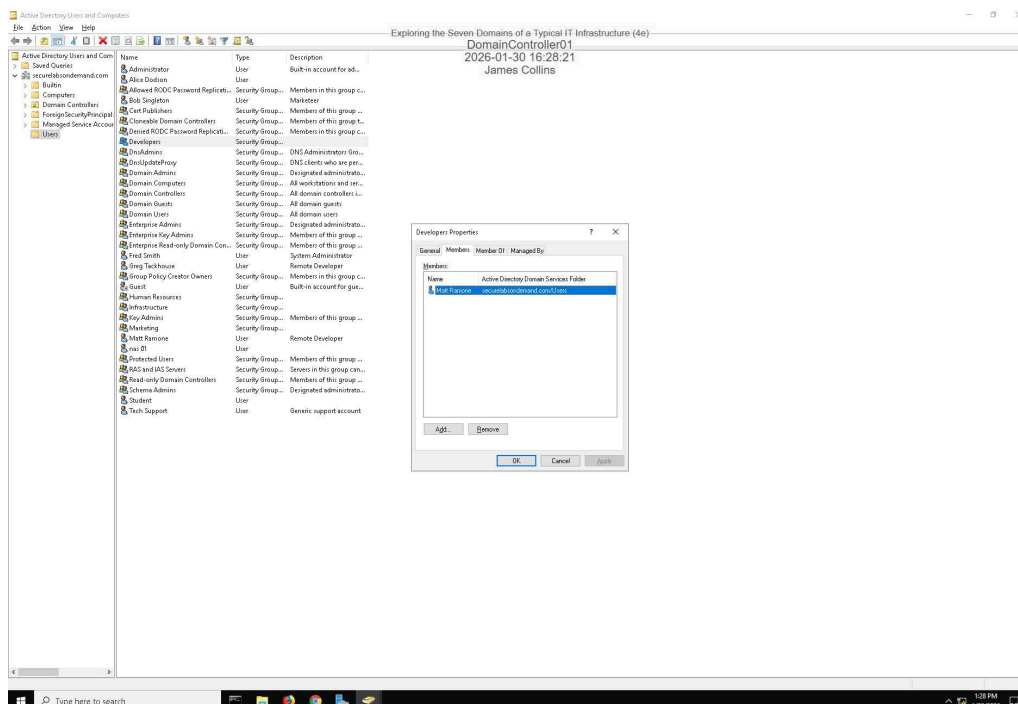
C:\Users\james>
```

Part 3: Explore the System/Application Domain

4. Make a screen capture showing the **whoami** results.

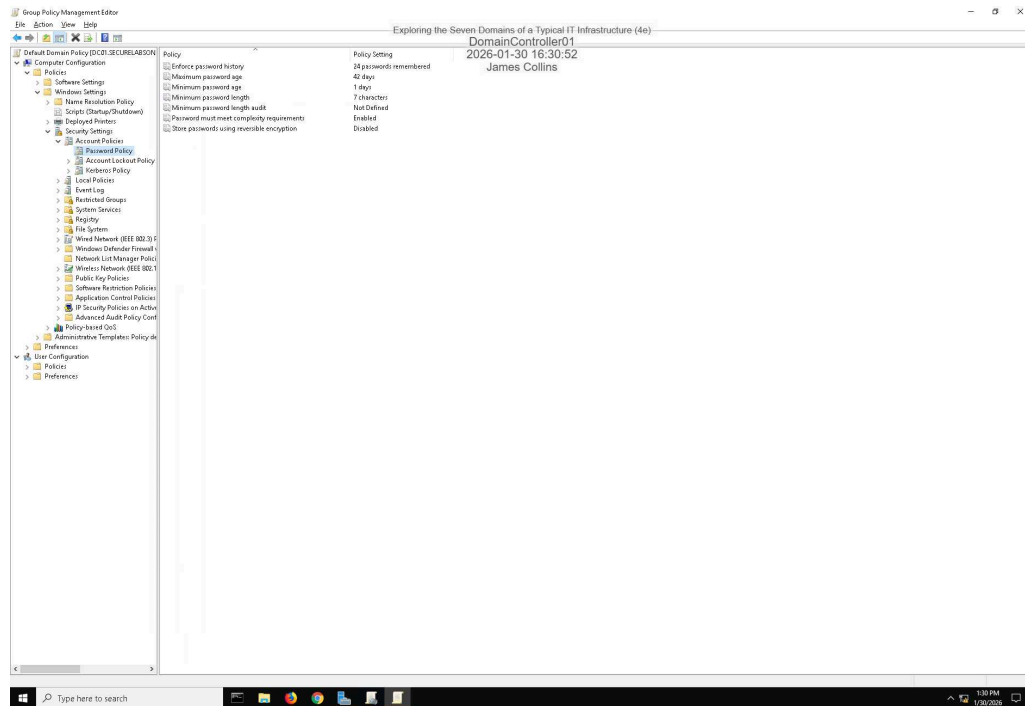


10. Make a screen capture showing the members of the Developers AD group.

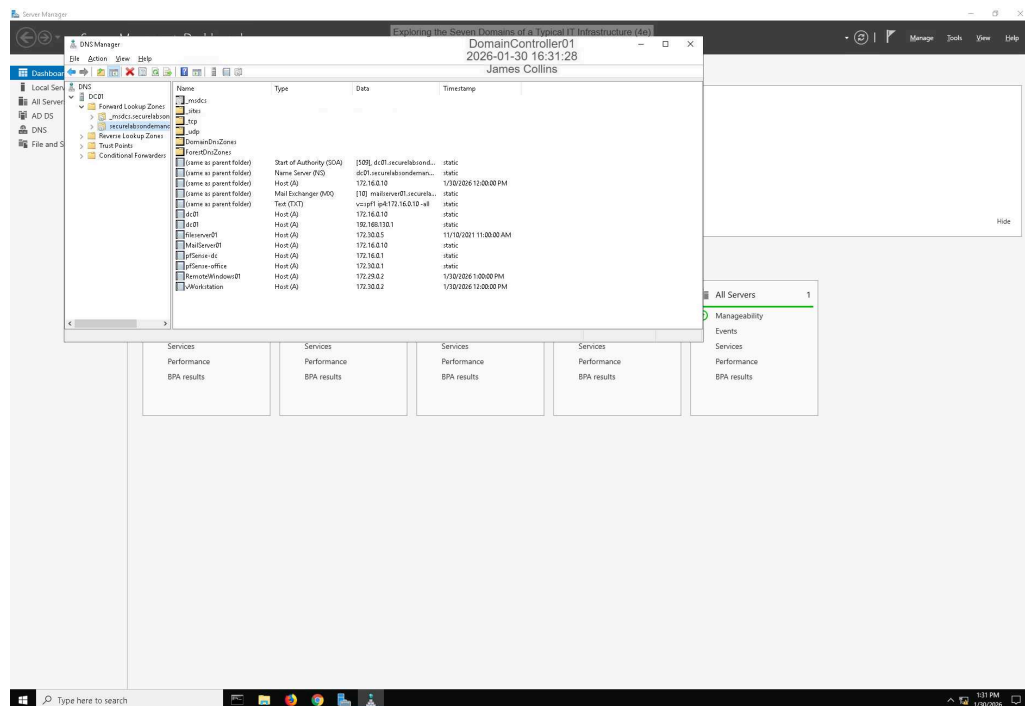


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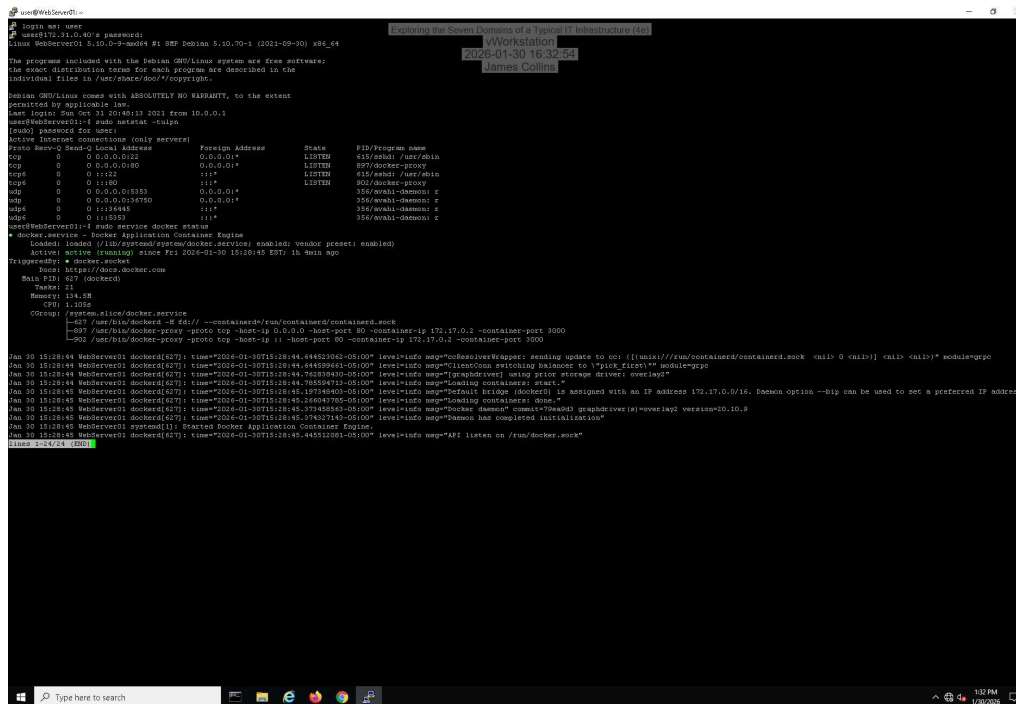
16. **Make a screen capture** showing the **password policy settings** in the **Group Policy Management Console**.



20. **Make a screen capture** showing the **DNS entries**.



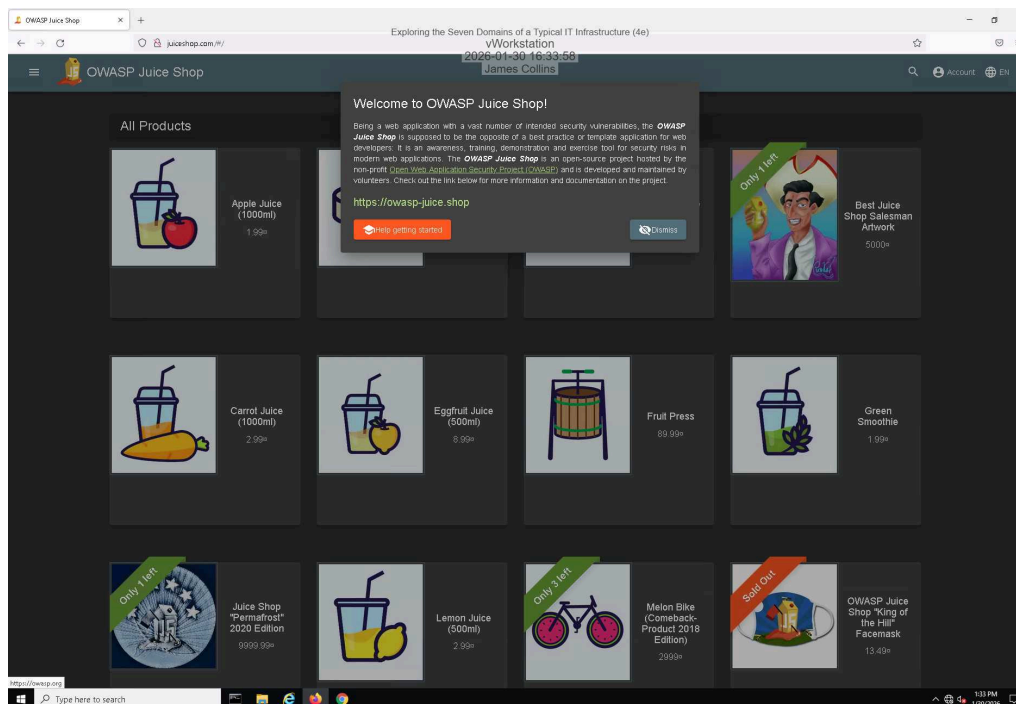
28. Make a screen capture showing the Docker service status.



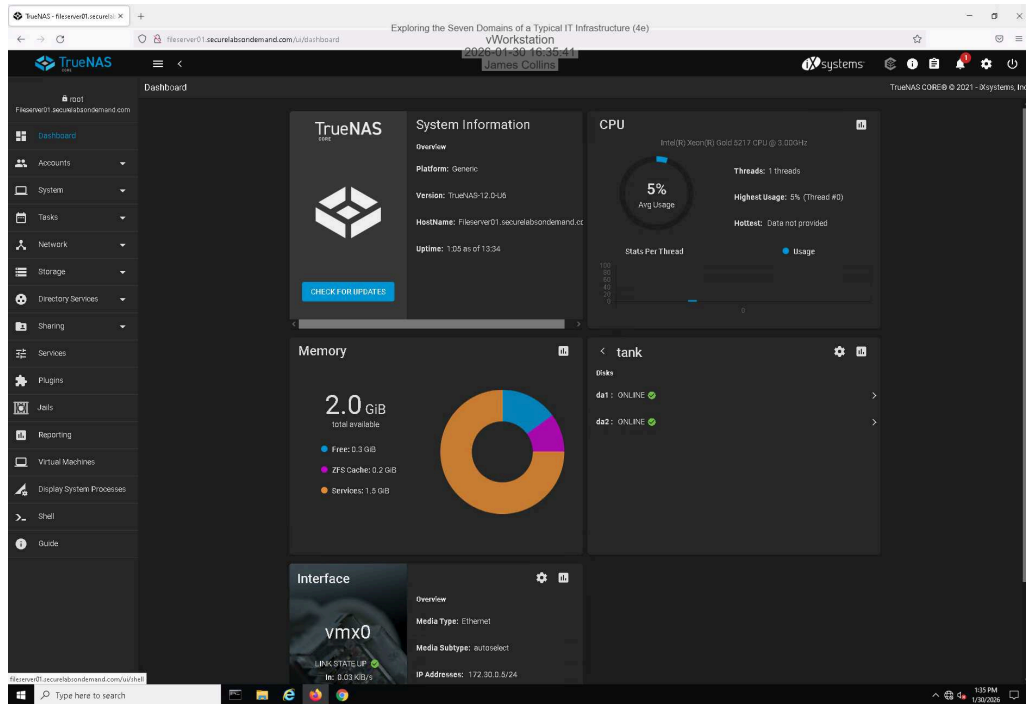
```
root@kali:~# docker service status
docker.service - Docker Application Container Engine
   Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2024-01-30 15:20:45 EST; 1h 4min ago
     Triples: 1
   Main PID: 627 (dockerd)
    Tasks: 21
    Memory: 144.5M
      CPU: 1.105s
    CGroup: /system.slice/docker.service
            └─┬─ /usr/bin/dockerd -s fd:// --containerd=/run/containerd/containerd.sock
               └─┬─ /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 80 --container-ip 172.17.0.2 --container-port 8080
                  └─ /usr/bin/docker-proxy -proto tcp -host-ip 11 -host-port 90 --container-ip 172.17.0.2 --container-port 9000

Jan 30 15:20:44 WebServer01 dockerd[627]: time="2024-01-30T15:20:44.64452042-05:00" level=info msg="codeResolverWapper: sending update to oci [[{"name":"/run/containerd/containerd.sock", "call":0,"callid":1}]] callid=1" module=grpc
Jan 30 15:20:44 WebServer01 dockerd[627]: time="2024-01-30T15:20:44.64459661-05:00" level=info msg="ClientConn switching balancer to \"pick_first\"" module=grpc
Jan 30 15:20:44 WebServer01 dockerd[627]: time="2024-01-30T15:20:44.76230430-05:00" level=info msg="[graphdriver] using prior storage driver: overlay2"
Jan 30 15:20:44 WebServer01 dockerd[627]: time="2024-01-30T15:20:44.76859473-05:00" level=info msg="Loading containers: start."
Jan 30 15:20:44 WebServer01 dockerd[627]: time="2024-01-30T15:20:44.87934800-05:00" level=info msg="Default bridge (docker0) is assigned with an IP address 172.17.0.0/16, daemon option --bip can be used to set a preferred IP address"
Jan 30 15:20:44 WebServer01 dockerd[627]: time="2024-01-30T15:20:45.26603700-05:00" level=info msg="Loading containers: done."
Jan 30 15:20:44 WebServer01 dockerd[627]: time="2024-01-30T15:20:45.374327143-05:00" level=info msg="[graphdriver] using prior storage driver: overlay2"
Jan 30 15:20:44 WebServer01 dockerd[627]: time="2024-01-30T15:20:45.374327143-05:00" level=info msg="Daemon has completed initialization"
Jan 30 15:20:45 WebServer01 dockerd[627]: time="2024-01-30T15:20:45.445522081-05:00" level=info msg="API listen on /run/docker.sock"
```

31. Make a screen capture showing the juiceshop.com web page.



36. Make a screen capture showing the disks in the tank volume.



Section 3: Challenge and Analysis

Part 1: Explore the User Domain

Based on your research, **identify** at least **two compelling threats** to the User Domain and **two effective security controls** used to protect it. Be sure to cite your sources.

Incomplete

Part 2: Research Additional Security Controls

Based on your research, **identify** security controls that could be implemented in the Workstation, LAN, LAN-to-WAN, WAN, Remote Access, and System/Application Domains. **Recommend** and **explain** one security control for each domain. Be sure to cite your sources.

Incomplete